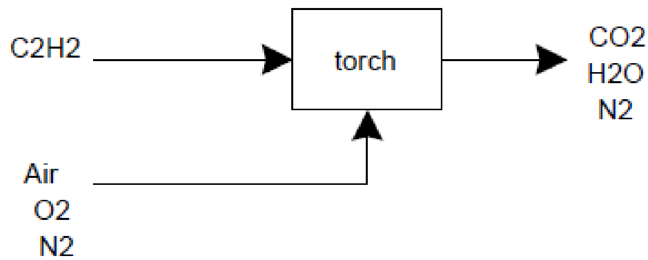
P6.47

Steps 1 and 2. Draw a diagram and define system.



Combustion of acetylene produces CO_2 and H_2O. Since oxygen is fed in stoichiometric quantity, and acetylene is completely combusted, no oxygen leaves the torch. The system is the torch and its contents.

Steps 3-5. Choose components, define stream variables, check units, define basis.
Components are acetylene (A), oxygen (O), nitrogen (N), carbon dioxide (C) and water (W). Stream variables will be defined as "in" or "out". We are not given a basis, so we'll choose one: $\dot{n}_{A,in} = 100$ gmol/h. We'll work in kJ and in gmol.

Step 6. Set up and solve process flows.
The balanced chemical reaction equation is

$$C_2H_2 + \frac{5}{2}O_2 \rightarrow 2CO_2 + H_2O$$

From the specification of stoichiometric feed ratio:

$$\dot{n}_{O,in} = 250 \text{ gmol/h}$$

Since air is 21 mol% O_2 and 79 mol% N_2:

$$\dot{n}_{N,in} = \left(\frac{79}{21}\right)250 = 940.5 \text{ gmol/h} = \dot{n}_{N,out}$$

From a steady-state balance on the torch, given the stoichiometry and 100% conversion,

$$\dot{n}_{C,out} = 200 \text{ gmol/h}$$
$$\dot{n}_{W,out} = 100 \text{ gmol/h}$$

Step 7. Define energy of process streams.
Only enthalpy changes are of interest. One choice for a reference state is the elements in their normal state of aggregation and at 298 K and 1 atm. This is convenient when we have a chemical reaction because we can then use enthalpy of formation data. Another convenient choice is the combustion products: CO_2(g), H_2O(g), and N_2(g) at 298 K and 1 atm. By choosing this reference state, we can use the enthalpy of combustion data. Let's work with the latter.

We'll assume the acetylene and air in the inlet stream are at 298 K (close to room temperature). From the data in Table B.3:

$$\hat{H}_{A,in} = -\Delta\hat{H}^{\circ}_{c,A} = 1257 \text{ kJ/gmol}$$
$$\hat{H}_{O,in} = 0 \text{ kJ/gmol}$$
$$\hat{H}_{N,in} = 0 \text{ kJ/gmol}$$

The outlet stream is at the adiabatic flame temperature T_f. The compounds in the stream are all in the gas phase, so they differ from the reference state only in temperature and we calculate the enthalpy at T_f by using C_p. We should really use the polynomial expressions but since this is just a relative calculation anyway (we are comparing acetylene relative to methane), we'll be very lazy and use approximate C_p's. We also neglect any enthalpy of mixing.

$$\hat{H}_{C,out} \approx C_p\left(T_f - T_{ref}\right) = \frac{37\left(T_f - 298\right)}{1000} = 0.037T_f - 11 \text{ kJ/gmol}$$
$$\hat{H}_{W,out} \approx C_p\left(T_f - T_{ref}\right) = \frac{33.6\left(T_f - 298\right)}{1000} = 0.0336T_f - 10 \text{ kJ/gmol}$$
$$\hat{H}_{N,out} \approx C_p\left(T_f - T_{ref}\right) = \frac{29.1\left(T_f - 298\right)}{1000} = 0.029T_f - 8.7 \text{ kJ/gmol}$$

Step 8. Identify heat and work.
This is an adiabatic flame, so there is no heat. Nor is there any work.

Step 9. Define changes in system energy
None - we assume steady state.

Steps 10-12: Write energy balance, solve, check.

$$\sum_{in}\dot{N}_j\hat{H}_j = \sum_{out}\dot{N}_j\hat{H}_j$$

$$(100)(1257)+250(0)+940.5(0)=200(0.037T_f-11)+100(0.0336T_f-10)+940.5(0.029T_f-8.7)$$

The solution is

$T_f \approx 3600\ K$ for acetylene

If we do the exact same calculation, but for methane (keeping in mind that the stoichiometry is different as well as the enthalpy of combustion, we find

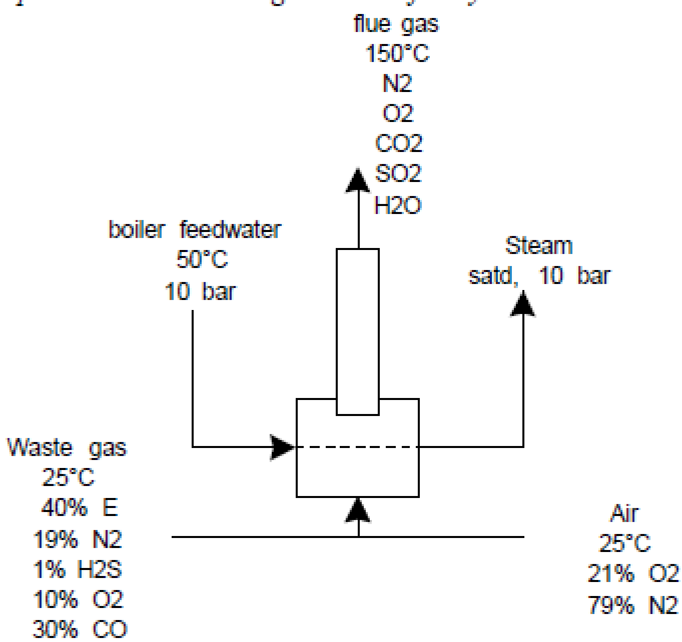
$T_f \approx 2800\ K$ for methane

or 800 K cooler.

The approximation of adiabatic operation of the torch is a poor one, as is the neglect of the temperature dependence of the heat capacity. Still, this shows that acetylene torches can reach much higher temperatures than can methane torches. This makes a huge difference, because of the high melting temperatures of many metals.

P6.48

Steps 1 and 2. Draw a diagram and define system.



Air and waste gas mix and are combusted in the firebox of the furnace. Boiler feedwater enters through tubes lining the wall of the furnace, where heat from the fire is transferred to the water, producing steam. We will choose two different systems: first we'll look at the contents of the firebox not including the tubes, and second we'll look at just the contents of the tubes. (Another appropriate choice is the entire contents of the firebox including the tubes as the system.)

Steps 3-5. Choose components, define stream variables, check units, define basis.
There are many components: oxygen (O), nitrogen (N), hydrogen sulfide (H), sulfur dioxide (S), water (W), carbon monoxide (CM), carbon dioxide (CD) and ethane (E). For units, we'll use gmol for the waste gas, air, and flue gas, and kg for the boiler feedwater and steam. (Mixing the units of mole and mass will not cause us problems, because the streams do NOT mix.) Streams will be defined as "waste", "air", "flue", "bfw" for boiler feedwater, and "steam" for steam. No basis is given, we'll choose 100 gmol/s of waste gas as a convenient basis:

$$\dot{n}_{waste} = 100 \text{ gmol/s}$$

Based on the composition of the waste gas stream, then

$$\dot{n}_{E,waste} = 40 \text{ gmol/s}$$

$$\dot{n}_{N,waste} = 19 \text{ gmol/s}$$

$$\dot{n}_{O,waste} = 10 \text{ gmol/s}$$

$$\dot{n}_{H,waste} = 1 \text{ gmol/s}$$

$$\dot{n}_{CM,waste} = 30 \text{ gmol/s}$$

Step 6. Set up and solve process flows.
Considering only the system of the firebox (excluding the steam/boiler feedwater), we have several reactions to consider:

$$C_2H_6 + \frac{7}{2}O_2 \rightarrow 2CO_2 + 3H_2O \qquad \text{(R1)}$$

$$CO + \frac{1}{2}O_2 \rightarrow CO_2 \qquad \text{(R2)}$$

$$H_2S + \frac{3}{2}O_2 \rightarrow SO_2 + H_2O \qquad \text{(R3)}$$

The stoichiometric flow rate of oxygen required for complete combustion is

$$\left(\frac{7}{2}\right)(40) + \left(\frac{1}{2}\right)(30) + \left(\frac{3}{2}\right)(1) - 10 = 146.5 \text{ gmol/s}$$

Since air is fed at 20% excess,

$$\dot{n}_{O,air} = 1.2 \times 146.5 = 175.8 \text{ gmol/s}$$

$$\dot{n}_{N,air} = \frac{79}{21} 175.8 = 661.3 \text{ gmol/s}$$

From material balance equations on ethane, carbon monoxide, and hydrogen sulfide (assuming complete combustion):

$$\dot{n}_{E,flue} = 0 = 40 - \dot{\xi}_1 \quad \rightarrow \quad \dot{\xi}_1 = 40 \text{ gmol/s}$$
$$\dot{n}_{CM,flue} = 0 = 30 - \dot{\xi}_2 \quad \rightarrow \quad \dot{\xi}_2 = 30 \text{ gmol/s}$$
$$\dot{n}_{H,flue} = 0 = 1 - \dot{\xi}_3 \quad \rightarrow \quad \dot{\xi}_3 = 1 \text{ gmol/s}$$

Finishing the material balance equations:

$$\dot{n}_{CD,flue} = 2\dot{\xi}_1 + \dot{\xi}_2 = 80 + 30 = 110 \text{ gmol/s}$$
$$\dot{n}_{S,flue} = \dot{\xi}_3 = 1 \text{ gmol/s}$$
$$\dot{n}_{W,flue} = 3\dot{\xi}_1 + \dot{\xi}_3 = 120 + 1 = 121 \text{ gmol/s}$$
$$\dot{n}_{N,flue} = \dot{n}_{N,waste} + \dot{n}_{N,air} = 19 + 661.3 = 680.3 \text{gmol/s}$$
$$\dot{n}_{O,flue} = \dot{n}_{O,waste} + \dot{n}_{O,air} - \frac{7}{2}\dot{\xi}_1 - \frac{1}{2}\dot{\xi}_2 - \frac{3}{2}\dot{\xi}_3 = 10 + 175.8 - 140 - 15 - 1.5 = 29.3 \text{ gmol/s}$$

Step 7. Define energy of process streams

We have some options. Option 1: we start with the waste gas and air at 298 K and 1 atm, calculate the enthalpy change associated with combustion at 298 K, then calculate the enthalpy change associated with increasing the temperature of the flue gases from 298 K to the outlet temperature, 150°C (423 K). Option 2: we start with the elements at 298 K and 1 atm, and calculate enthalpies of compounds at 298 K from enthalpy of formation data, then correct flue gases for the temperature increase. Either option is fine; we'll go with option 1 here.

Here is the useful data (all for water in the vapor phase as the product of combustion)

Ethane:	$\Delta H_c^\circ = -1428.6$ kJ/gmol
CO:	$\Delta H_c^\circ = -283$ kJ/gmol
H_2S:	$\Delta H_c^\circ = -518.7$ kJ/gmol

CO_2:	$C_p = 19.80 + 0.07344T - 5.602 \times 10^{-5}T^2 + 1.7115 \times 10^{-8}T^3$
H_2O:	$C_p = 32.24 + 0.001924T + 1.055 \times 10^{-5}T^2 - 3.569 \times 10^{-9}T^3$
O_2:	$C_p = 29.1 + 0.01158T - 6.076 \times 10^{-6}T^2 + 1.311 \times 10^{-8}T^3$
N_2:	$C_p = 31.15 - 0.01357T + 2.680 \times 10^{-5}T^2 - 1.168 \times 10^{-8}T^3$
SO_2:	$C_p = 23.85 + 0.06699T - 4.961 \times 10^{-5}T^2 + 1.328 \times 10^{-8}T^3$

$$H_{flue} = \sum_k \dot{\xi}_k \Delta\hat{H}_{c,k} + \sum_i \dot{n}_{i,flue} \int_{T_{in}}^{T_{out}} C_{p,i}\,dT$$
$$H_{flue} = \left(40(-1428.6)\right) + \left(30(-283)\right) + \left(1(-518.7)\right)$$
$$+ \left(110(4.97)\right) + \left(121(4.72)\right) + \left(29.3(3.73)\right) + \left(680.3(3.65)\right) + \left(1(5.27)\right)$$
$$H_{flue} = -62{,}400 \text{ kJ/s}$$

Step 8. Identify heat and work.
There is heat that leaves this system (and is transferred to the steam in the tubes), but no work.

Step 9. Define changes in system energy
None – assumed steady state.

Steps 10-12: Write energy balance, solve, check.

$$\dot{H}_{out} - \dot{H}_{in} = \dot{Q} = -62{,}400 \text{ kJ/s}$$

The heat that leaves this system is transferred to the contents of the tubes. Now we change the system to the tubes and their contents. The energy balance is

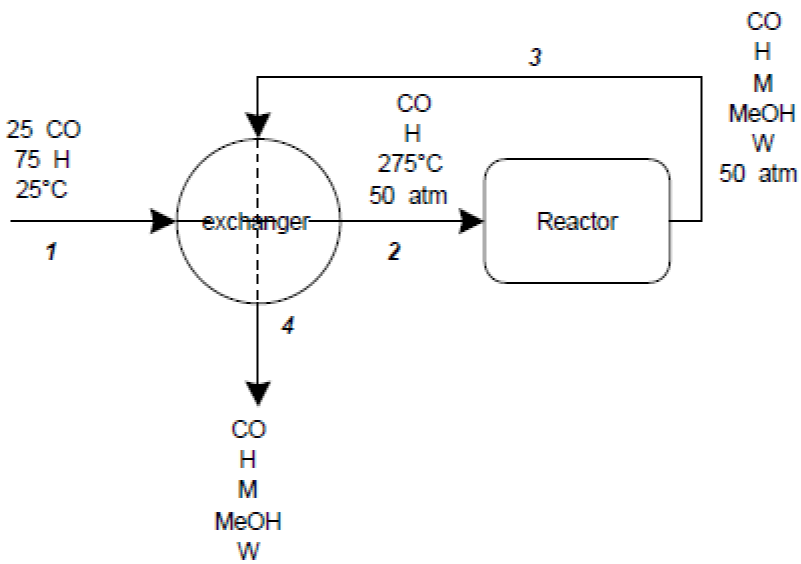
$$\dot{H}_{out} - \dot{H}_{in} = \dot{Q} = +62{,}400 \text{ kJ/s}$$

We can choose a different reference state for the boiler feedwater and steam (because they are in a different system.) Let's use the triple point of water and then get our data from the steam tables.

$$\dot{m}_{steam}\left(\hat{H}_{steam} - \hat{H}_{bfw}\right) = \dot{m}_{steam}(2777.1 - 210.19) = +62{,}400 \text{ kJ/s}$$
$$\dot{m}_{steam} = 24.3 \text{ kg/s}$$

Since the basis was 100 gmol/s waste gas, we calculate that we produce 0.243 kg steam per gmol waste gas burned.

P6.68
The labeled flow diagram for the part of the process of interest is



First we complete process flow calculations on the reactor. Flows are all in gmol/s.

$$\dot{n}_{CO,2} = 25$$
$$\dot{n}_{H,2} = 75$$
$$\dot{n}_{MeOH,3} = 18.7 = \dot{\xi}_1$$

$$\dot{n}_{CO,3} - \dot{n}_{CO,2} = 0.8 \times 25 = 20 = \dot{\xi}_1 + \dot{\xi}_2 = 18.7 + \dot{\xi}_2$$
$$\dot{\xi}_2 = 1.3$$
$$\dot{n}_{H,3} = 75 - 2(18.7) - 3(1.3) = 33.7$$
$$\dot{n}_{M,3} = \dot{\xi}_2 = 1.3$$
$$\dot{n}_{W,3} = \dot{\xi}_2 = 1.3$$

Flows in stream 4 equal those in stream 3.

Now we turn to the energy balance. The reactor is adiabatic and operates at steady state, and there is no work term. The differential energy balance reduces to:

$$\dot{H}_2 = \dot{H}_3$$

To calculate the difference in enthalpy between streams 2 and 3, we construct a path:

(a) decrease pressure of stream 2 from 50 atm to 1 atm
(b) decrease temperature of stream 2 from 275°C to 25°C
(c) react stream 2 to stream 3 through reactions R1 and R2 at 25°C
(d) increase temperature of stream 3 from 25°C to Tout
(e) increase pressure of stream 3 from 1 atm to 50 atm

The enthalpy change associated with change in pressure is negligible (steps (a) and (e)). (We are also neglecting any change in enthalpy due to mixing.) For the remaining calculations, we collect the following data (all compounds in vapor phase):

$$C_{p,CO} = 30.87 - 0.01285T + 2.789\times10^{-5}T^2 - 1.272\times10^{-8}T^3$$

$$C_{p,H} = 27.14 + 0.0093T - 1.381\times10^{-5}T^2 + 7.645\times10^{-9}T^3$$

$$C_{p,MeOH} = 21.15 + 0.07092T + 2.587\times10^{-5}T^2 - 2.852\times10^{-8}T^3$$

$$C_{p,M} = 19.25 + 0.05213T + 1.197\times10^{-5}T^2 - 1.132\times10^{-8}T^3$$

$$C_{p,W} = 32.24 - 0.001924T + 1.055\times10^{-5}T^2 - 3.596\times10^{-9}T^3$$

$$\Delta\hat{H}^\circ_{r1} = (-1)(-110.53) + (-2)(0) + (1)(-200.94) = -90.41 \text{ kJ/gmol}$$

$$\Delta\hat{H}^\circ_{r2} = (-1)(-110.53) + (-3)(0) + (1)(-74.52) + (1)(-241.83) = -205.82 \text{ kJ/gmol}$$

Both reactions are exothermic, which makes sense given that the reactor effluent is used to heat the reactants fed to the reactor.

$$\dot{H}_3 - \dot{H}_2 = \sum_i \dot{n}_{i2}\int_{275}^{25} C_{p,i}\,dT + \sum_k \dot{\xi}_k \Delta\hat{H}^\circ_{rk} + \sum_i \dot{n}_{i3}\int_{25}^{T_3} C_{p,i}\,dT = 0$$

We know everything but T_3 so can solve for the reactor effluent temperature(using Solver on a spreadsheet):

$$T_3 = 1300 \text{ K} \cong 1020°C$$

The temperature of the product stream leaving the exchanger can be calculated either by choosing the exchanger as the system, or the exchanger plus reactor. We'll use just the exchanger. Assuming the heat exchanger is well-insulated, the steady state energy balance simplifies to:

$$\dot{H}_2 - \dot{H}_1 = \dot{H}_3 - \dot{H}_4$$

The only change in enthalpy occurs due to a change in temperature, so the energy balance equation simplifies further to:

$$\sum_i n_{i2}\int_{25}^{275} C_{pi}\,dT = \sum_i n_{i3}\int_{T_4}^{1000} C_{pi}\,dT$$

We find the product outlet temperature that satisfies this equation:

$$T_3 = 1060 \text{ K} \cong 790°C$$

The temperature of the reactor effluent drops by 230°C. This is reasonable, since the reactant stream increased by about the same amount - 250°C – by heat exchange with the effluent.